OSPF

Routing Protocols and Concepts – Chapter 11

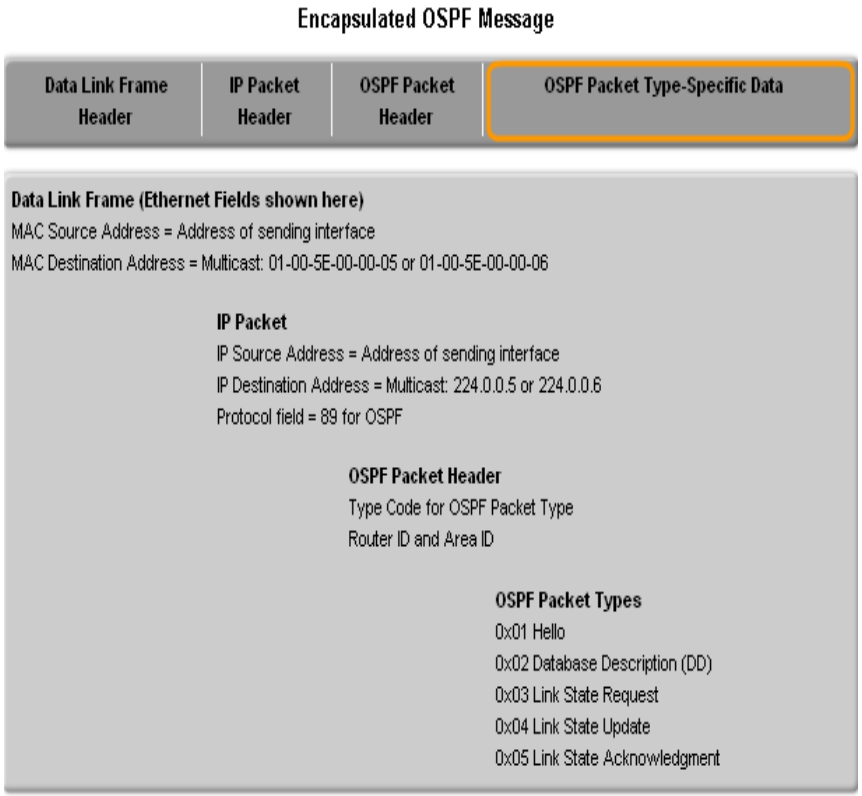
Cisco | Networking Academy®
Mind Wide Open™

Version 4.0

© 2007 Cisco Systems, Inc. All rights reserved. Cisco Public

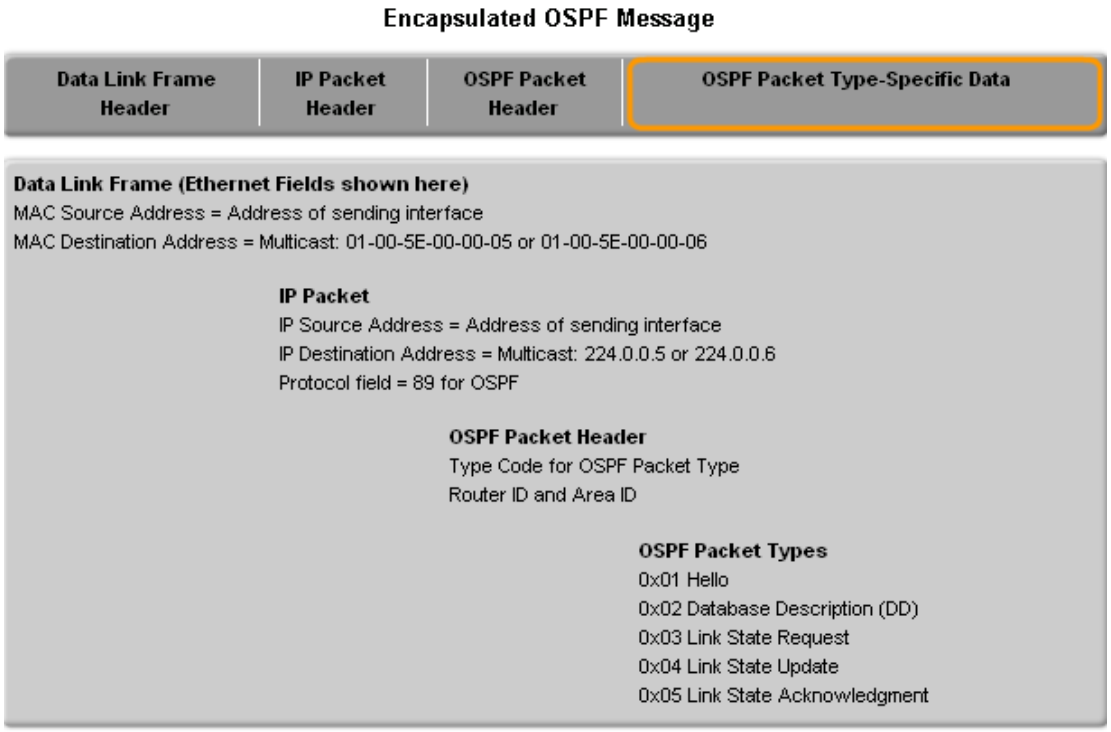
Introduction to OSPF

- OSPF Message Encapsulation
  - OSPF packet type
    - There exist 5 types
  - OSPF packet header
    - Contains - Router ID and area ID and Type code for OSPF packet type
  - IP packet header
    - Contains - Source IP address, Destination IP address, & Protocol field set to 89



Introduction to OSPF

- OSPF Message Encapsulation
  - Data link frame header
  - Contains - Source MAC address and Destination MAC address



Introduction to OSPF

- OSPF Packet Types

Type	Packet Name	Description
1	Hello	Discovers neighbors and builds adjacencies between them
2	Database Description (DBD)	Checks for database synchronization between routers
3	Link-State Request (LSR)	Requests specific link-state records from router to router
4	Link-State Update (LSU)	Sends specifically requested link-state records
5	Link-State Acknowledgement (LSAck)	Acknowledges the other packet types

Introduction to OSPF

- OSPF Link-state Updates
  - Purpose of a Link State Update (LSU)
    - Used to deliver link state advertisements
  - Purpose of a Link State Advertisement (LSA)
    - Contains information about neighbors & path costs

LSUs Contain Link-State Advertisements (LSAs)

Type	Packet Name	Description
1	Hello	Discovers neighbors and builds adjacencies between them
2	DBD	Checks for database synchronization between router
3	LSR	Requests specific link-state records from router to router
4	LSU	Sends specifically requested link-state records
5	LSAck	Acknowledges the other packet types

The acronyms LSA and LSU are often used interchangeably.

An LSU contains one or more LSAs.

LSAs contain route information for destination networks.

LSA specifics are discussed in CCNP.

LSA Type	Description
1	Router LSAs
2	Network LSAs
3 or 4	Summary LSAs
5	Autonomous System Extrenal LSAs
6	Multicast OSPF LSAs
7	Defined for Not-So-Stubby Areas
8	External Attributes LSA for Border Gatway Protocol(BGP)
9,10,11	Opaque LSAs

OSPF

- Open Shortest Path First (OSPF) adalah link-state routing protokol yang dikembangkan sebagai pengganti untuk distance vector routing protocol RIP.
- RIP merupakan routing protocol yang digunakan pada awal perkembangan jaringan dan Internet, tetapi ketergantungan pada hop sebagai satu-satunya ukuran untuk memilih rute terbaik dan cepat menjadi tidak dapat diterima dalam jaringan yang lebih besar yang membutuhkan solusi routing lebih kuat.
- OSPF adalah protokol routing tanpa kelas (*classless) yang menggunakan konsep area untuk skalabilitas.*
- Keunggulan utama OSPF setelah berakhirnya RIP adalah konvergensi yang cepat dan skalabilitas untuk implementasi jaringan yang jauh lebih besar.
- OSPF adalah protokol routing yang menggunakan dasar linkstate. Linkstate menggunakan algoritma SPF (Shortest Path First) untuk mempopulasikan routing table. OSPF berbagi informasi routing dengan antar router.
- Dalam penggunaan protokol routing OSPF, perlu didefinisikan terlebih dahulu “area”.

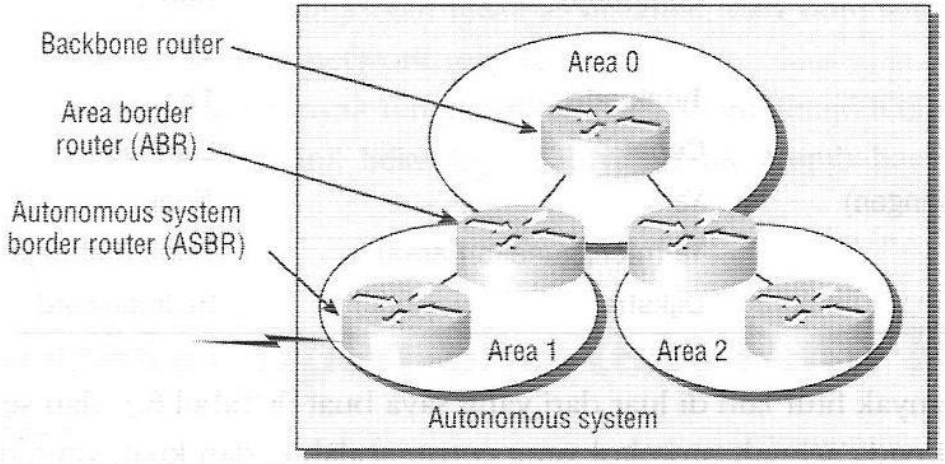
OSPF

- OSPF (**Open Shortest Path First**) adalah sebuah routing protokol standard terbuka yang telah diimplementasikan oleh sejumlah besar vendor jaringan, termasuk Cisco.
- Jika kita memiliki banyak router dan tidak semuanya adalah Cisco, maka kita tidak dapat menggunakan EIGRP. Jadi pilihan kita tinggal RIPv1, RIPv2, atau OSPF.
- Jika network kita adalah network besar, maka pilihan kita satu-satunya hanya OSPF atau sesuatu yang disebut Route Redistribution--sebuah layanan penerjemahan antar-routing protocol.

Dasar-Dasar OSPF

- OSPF bekerja dengan sebuah algoritma yang disebut Algoritma Dijkstra
  Pertama, sebuah pohon jalur terpendek akan dibangun, kemudian
  Routing table akan diisi dengan jalur-2 terbaik yang dihasilkan oleh pohon tersebut.
- OSPF melakukan converge dengan cepat, meskipun tidak sesepat EIGRP dan OSPF mendukung multiple route dengan cost yang sama, ke tujuan yang sama.
- OSPF hanya mendukung routing IP
- OSPF adalah routing protocol jenis link-state pertama yang diperkenalkan.
- OSPF seharusnya dirancang dengan cara hirarkis (memisahkan internetwork yang besar menjadi internetwork-2 kecil yang disebut area). Ini adalah rancangan terbaik untuk OSPF. Alasannya:
  - Untuk mengurangi overhead (waktu pemrosesan) routing
  - Untuk mempercepat convergence
  - Untuk membatasi ketidakstabilan network di sebuah area dari network saja

CONTOH RANCANGAN OSPF



- Perhatikan gambar, setiap router terhubung ke backbone—yang disebut area 0

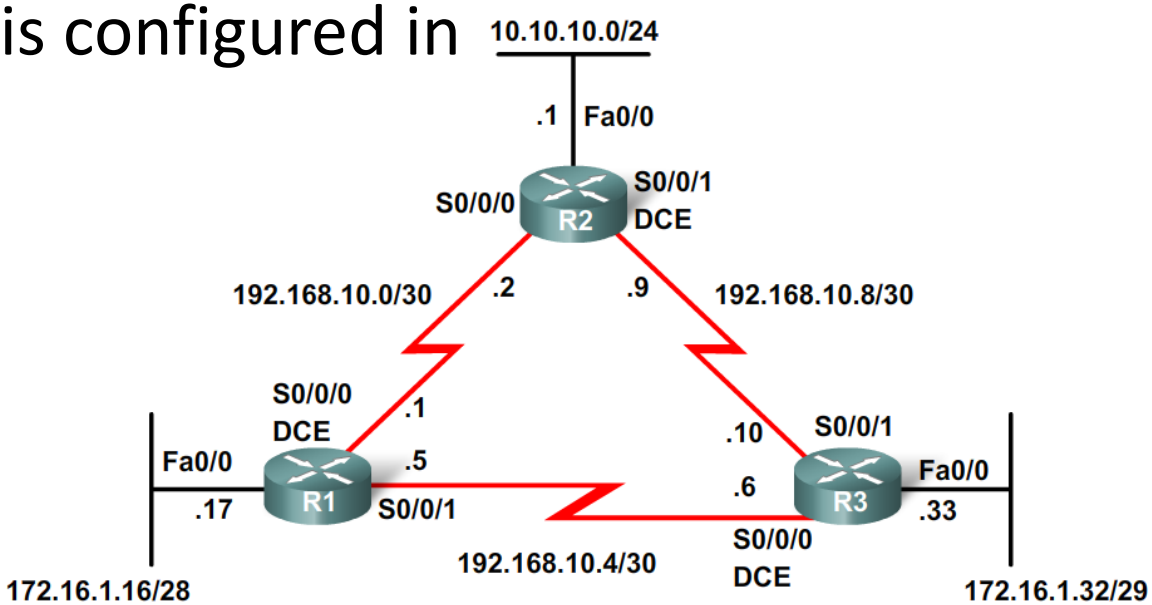
  OSPF harus memiliki sebuah area 0, dan semua router harus terhubung ke area ini jika memungkinkan.
- Router-2 yang menghubungkan area-2 lain ke backbone di dalam sebuah AS disebut Area Border Routers (ABRs). Meskipun demikian, paling sedikit satu interface harus berada di area 0
- OSPF bekerja didalam sebuah Autonomous System (AS), tetapi dapat juga menghubungkan banyak AS bersama. Router yang menghubungkan beberapa AS bersama disebut sebuah Autonomous System Boundary Router (ASBR)

PERHITUNGAN POHON SPF (SPF TREE)

- Dalam sebuah area, setiap router menghitung jalur terbaik/terpendek ke semua network di area yang sama. Perhitungan ini didasarkan pada informasi yang dikumpulkan di topology database dan sebuah algoritma yang disebut Algoritma Shortest Path First (SPF)
- Jika sebuah router memiliki interface-2 dibanyak area, maka pohon-2 yang terpisah akan dibuat untuk setiap area.
- Satu dari kriteria utama yang diperhitungkan selama proses pemilihan route dari algoritma SPF adalah metric atau cost dari setiap jalur yang potensial ke sebuah network. Tetapi perhitungan SPF ini tidak berlaku untuk route-2 yang berasal dari area lain.
- OSPF menggunakan sebuah metric yang disebut sebagai cost (biaya).
  - Sebuah cost dihitung untuk semua outgoing interface (interface yang mengirimkan data) yang termasuk dalam sebuah pohon SPF.
  - Cost dari keseluruhan jalur adalah penjumlahan dari cost-cost yang dimiliki oleh interface-2 outgoing disepanjang jalur tersebut.
  - Cisco menghitung Cost dgn rumus = 10^8 / bandwidth interface.
  - Nilai cost dapat dimanipulasi dalam range 1 s/d 65.535, perintahnya: ip ospf cost

Basic OSPF Configuration

- Lab Topology
- Topology used for this chapter
  - Discontiguous IP addressing scheme
  - Since OSPF is a classless routing protocol the subnet mask is configured in



Basic OSPF Configuration

- The router ospf command
- To enable OSPF on a router use the following command
  - R1(config)#**router ospf *process-id***
  - Process id
    - A locally significant number between **1** and **65535**
    - This means it does not have to match other OSPF routers

```
R1(config)#router ospf 1
R1(config-router)#
```

```
R2(config)#router ospf 1
R2(config-router)#
```

```
R3(config)#router ospf 1
R3(config-router)#
```

Basic OSPF Configuration

- OSPF network command
  - Requires entering:
    - **network address**
    - **wildcard mask** - the inverse of the subnet mask
    - **area-id -** area-id refers to the OSPF area – OSPF area is a group of routers that share link state information
  - Example: Router(config-router)#**network** network-address wildcard-ask **area** area-id

```
R1(config)#router ospf 1
R1(config-router)#network 172.16.1.16 0.0.0.15 area 0
R1(config-router)#network 192.168.10.0 0.0.0.3 area 0
R1(config-router)#network 192.168.10.4 0.0.0.3 area 0
```

```
R2(config)#router ospf 1
R2(config-router)#network 10.10.10.0 0.0.0.255 area 0
R2(config-router)#network 192.168.10.0 0.0.0.3 area 0
R2(config-router)#network 192.168.10.8 0.0.0.3 area 0
```

Basic OSPF Configuration

- Router ID
  - This is an IP address used to identify a router
  - 3 criteria for deriving the router ID
    - Use IP address configured with OSPF *router-id* command
      - Takes precedence over loopback and physical interface addresses
    - If router-id command not used then router chooses highest IP address of any loopback interfaces
    - If no loopback interfaces are configured then the highest IP address on any active interface is used

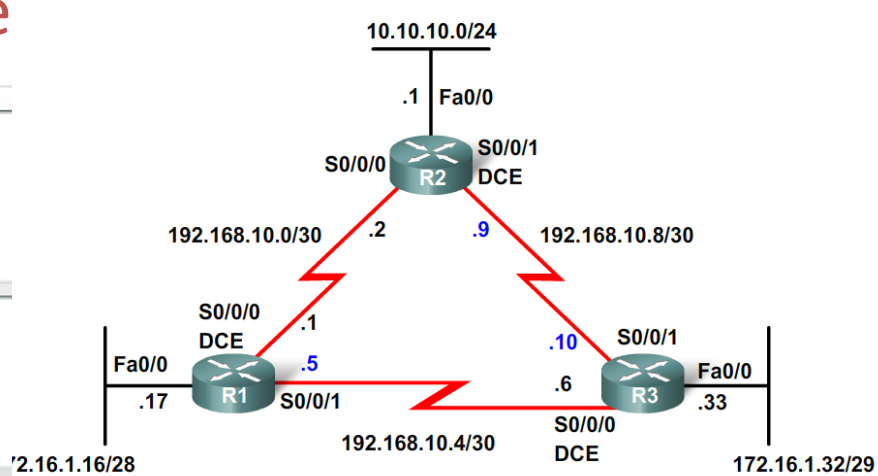
Basic OSPF Configuration

- OSPF Router ID
- Commands used to verify current router ID
  - Show ip protocols
  - Show ip ospf
  - Show ip ospf interface

```
R1#show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.10.5
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
R2#show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.10.9
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
R3#show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.10.10
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```



Basic OSPF Configuration

- Verifying OSPF - Additional Commands

Command	Description
Show ip protocols	Displays OSPF process ID, router ID, networks router is advertising & administrative distance
Show ip ospf	Displays OSPF process ID, router ID, OSPF area information & the last time SPF algorithm calculated
Show ip ospf interface	Displays hello interval and dead interval

OSPF

- OSPF adalah protokol routing yang menggunakan dasar linkstate. Linkstate menggunakan algoritma SPF (Shortest Path First) untuk mempopulasikan routing table. OSPF berbagi informasi routing dengan antar router.

Area

- Dalam penggunaan protokol routing OSPF, perlu didefinisikan terlebih dahulu “area”. Dimana untuk daerah jaringan backbone harus menggunakan area 0 atau 0.0.0.0
- Netmask dan Wildcard
- wildcard adalah kebalikan dari netmask.
- Contoh : netmask 255.255.0.0 berarti wildcardnya 0.0.255.255

Contoh OSPF

- Cisco#conf t
- Cisco(config)#router ospf 1
- cisco (config-router)#
- network 10.252.101.0 0.0.0.255 area 0

Keterangan:

- Untuk memulai “router ospf 1” dimana 1 adalah ID dari router tersebut
- Kemudian masukkan jaringannya dengan perintah “network 10.252.101.0 0.0.0.255 area 0”, yg artinya mendaftarkan network 10.252.101.0/24 di area 0 (backbone)

Mengkonfigurasi OSPF (1)

- Konfigurasi mendasar OSPF tidak sesederhana RIP, IGRP, dan EIGRP. Konfigurasi ini bisa menjadi benar-benar rumit ketika banyak opsi yang diperbolehkan di OSPF, ikut diperhitungkan.
- Ada dua unsur dasar konfigurasi OSPF:
  - Mengaktifkan OSPF
  - Mengkonfigurasi area-area OSPF
- Contoh Configurasi OSPF:

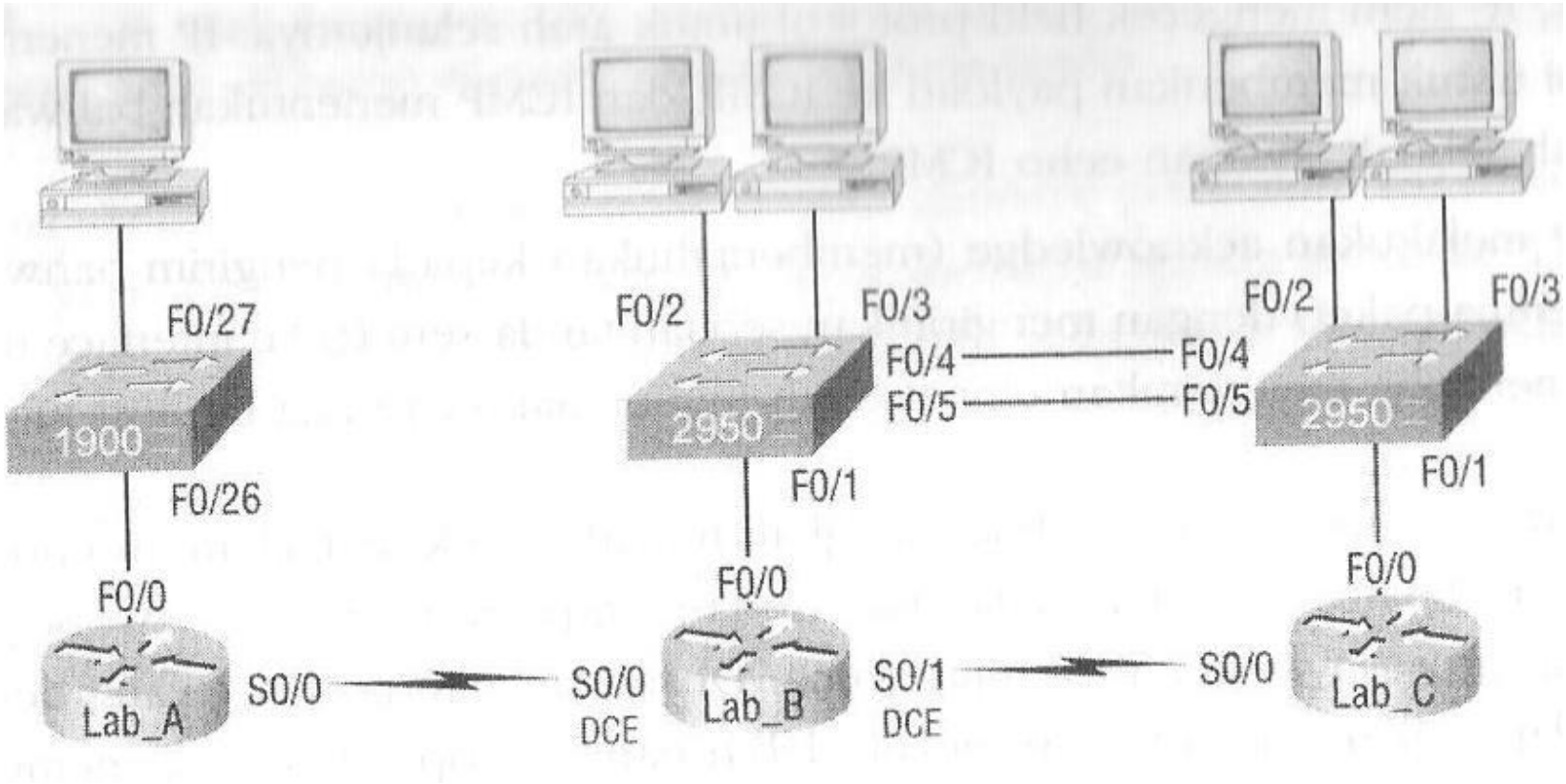
```
Lab_A#config terminal
Lab_A(config)#no router eigrp 10
Lab_A(config)#no router igrp 10
Lab_A(config)#no router rip
Lab_A(config)#router ospf 64999
Lab_A(config-router)#network 192.168.40.0 0.0.0.255 area 0
Lab_A(config-router)#network 192.168.50.0 0.0.0.255 area 0
Lab_A(config-router)#^Z
Lab_A#sh ip route
```

Mengkonfigurasi OSPF (2)

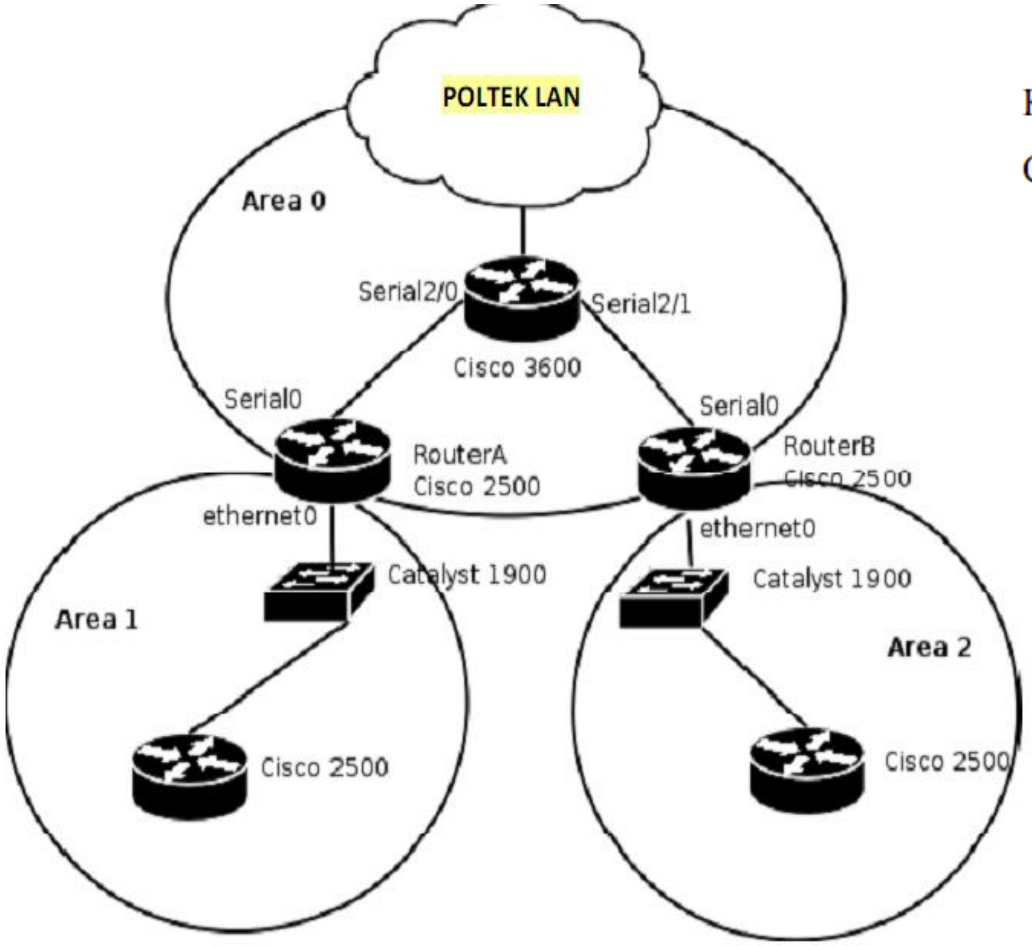
- Contoh perintah show pada OSPF:

Lab_A#sho ip ospf	menampilkan informasi OSPF yang bekerja pada router
Lab_A#sho ip ospf database	menampilkan informasi nomor dari link-link dan ID dari router tetangga serta database topologi
Lab_A#sho ip ospf interface	menampilkan semua informasi OSPF yang berkaitan dengan interface
Lab_A#sho ip ospf neighbor	melakukan summarize terhadap informasi OSPF yang relevan mengenai neighbors dan status adjacency
Lab_A#sho ip protocols	menampilkan informasi routing protocol apa yang dijalankan (OSPF, EIGRP, IGRP, atau RIP atau yang lainnya)

Latihan OSPF



Latihan OSPF



Keterangan Topologi

Cisco 3600

Serial 2/0 192.168.10.1/255.255.255.0

Serial 2/1 192.168.11.1/255.255.255.0

Router A

Serial0 192.168.10.2/255.255.255.0

ethernet0 172.13.5.1/255.255.0.0

Router B

Serial0 192.168.11.2/255.255.255.0

ethernet0 172.14.6.1/255.255.0.0

Dimana terdapat 3 area yang berbeda

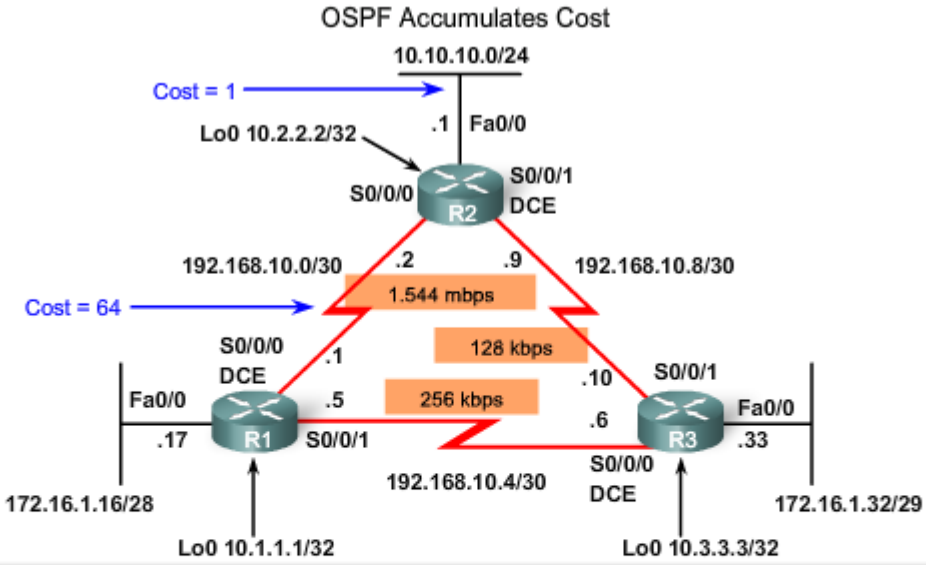
OSPF Metric

- OSPF uses **cost** as the metric for determining the best route
  - The best route will have the lowest cost
  - **Cost** is based on bandwidth of an interface
    - Cost is calculated using the formula
      - 10^8 / bandwidth
  - Reference bandwidth
    - Defaults to 100Mbps
    - Can be modified using
    - **Auto-cost reference-bandwidth** command

Interface Type	10^8/bps = Cost
Fast Ethernet and faster	10^8/100,000,000 bps = 1
Ethernet	10^8/10,000,000 bps = 10
E1	10^8/2,048,000 bps = 48
T1	10^8/1,544,000 bps = 64
128 kbps	10^8/128,000 bps = 781
64 kbps	10^8/64,000 bps = 1562
56 kbps	10^8/56,000 bps = 1785

OSPF Metric

- COST of an OSPF route
  - Is the accumulated value from one router to the next



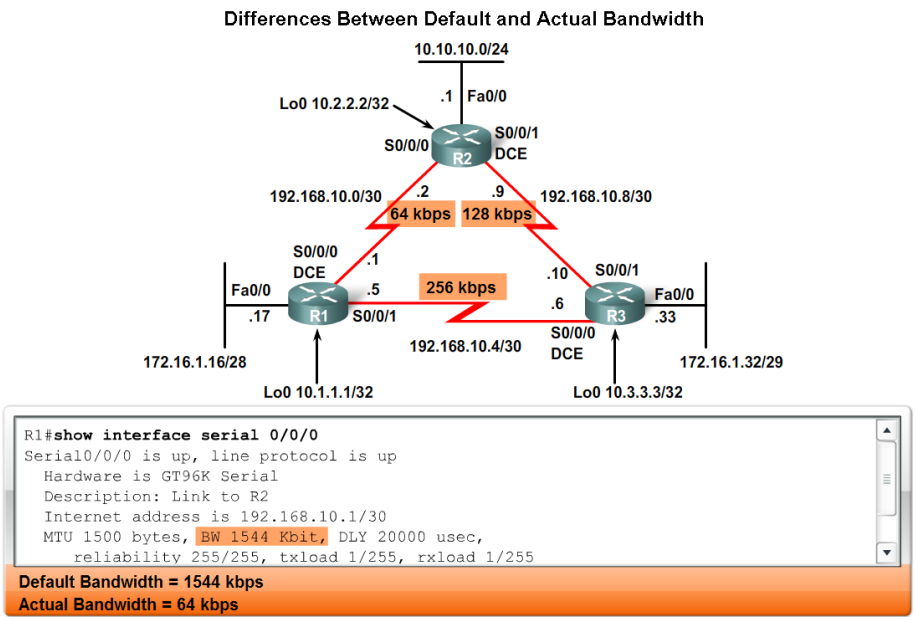

```
R1#show ip route
Codes: <some code output omitted>
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

***output omitted***
O       10.10.10.0/24 [110/65] via 192.168.10.2, 14:27:57, Serial0/0/0
```

Accumulated Cost = 65

OSPF Metric

- Usually the actual speed of a link **is different** than the default bandwidth
  - This makes it imperative that the bandwidth value reflects link's actual speed
    - Reason: so routing table has best path information
- The **show interface** command will display interface's bandwidth
  - Most serial link default to 1.544Mbps



Basic OSPF Configuration

- Modifying the Cost of a link
- Both sides of a serial link should be configured with the same bandwidth
  - Commands used to modify bandwidth value
    - **Bandwidth** command
      - Example: Router(config-if)#**bandwidth***bandwidth-kbps*
    - **ip ospf cost** command – allows you to directly specify interface cost
      - Example: R1(config)#interface serial 0/0/0
      - R1(config-if)#ip ospf cost 1562

```
R1(config)#inter serial 0/0/0
R1(config-if)#bandwidth 64
R1(config-if)#inter serial 0/0/1
R1(config-if)#bandwidth 256
R1(config-if)#end
R1#show ip ospf interface serial 0/0/0
Serial0/0 is up, line protocol is up
  Internet Address 192.168.10.1/30, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type POINT_TO_POINT, Cost: 1562
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  <output omitted>
```

10^8/64,000 bps = 1562

```
R1(config)#inter serial 0/0/0
R1(config-if)#ip ospf cost 1562
R1(config-if)#end
R1#show ip ospf interface serial 0/0/0
Serial0/0 is up, line protocol is up
  Internet Address 192.168.10.1/30, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type POINT_TO_POINT, Cost: 1562
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  <output omitted>
```

No Calculation Needed

Basic OSPF Configuration

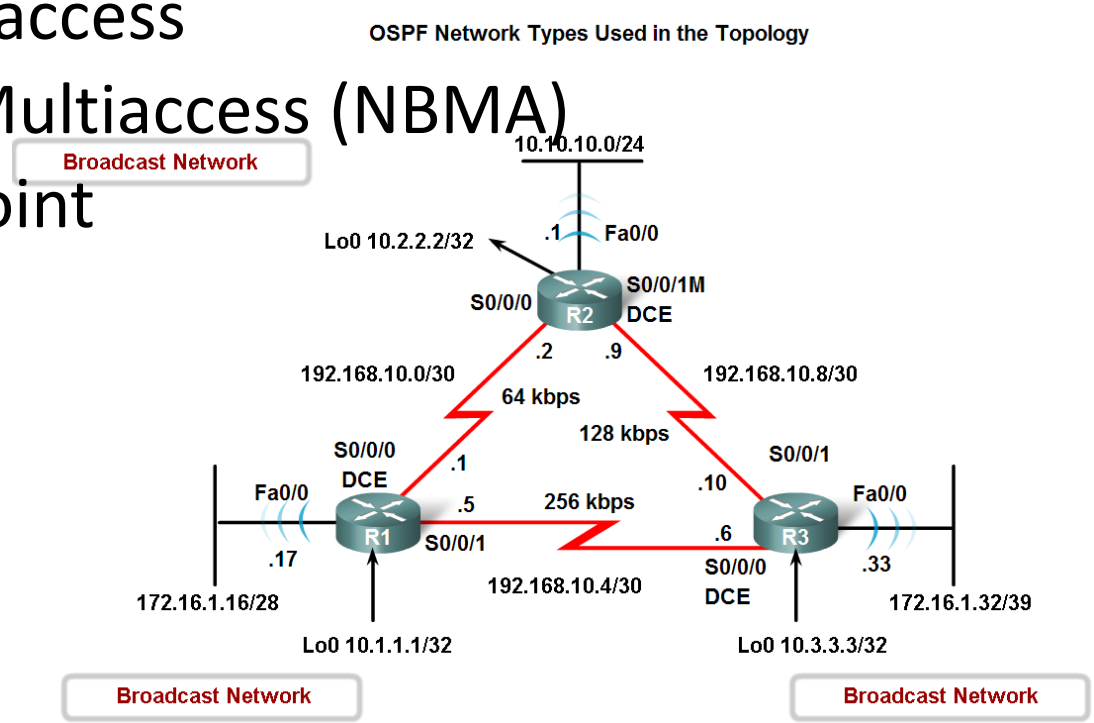
- Modifying the Cost of the link
- Difference between bandwidth command & the ip ospf cost command
  - **Ip ospf cost** command
    - Sets cost to a specific value
  - **Bandwidth** command
    - Link cost is calculated

Equivalent Commands

bandwidth Commands		ip ospf cost Commands
Router R1 R1(config)#interface serial 0/0/0 R1(config-if)#bandwidth 64	=	Router R1 R1(config)#interface serial 0/0/0 R1(config-if)#ip ospf cost 1562
R1(config)#interface serial 0/0/1 R1(config-if)#bandwidth 256	=	R1(config)#interface serial 0/0/1 R1(config-if)#ip ospf cost 390
Router R2 R2(config)#interface serial 0/0/0 R2(config-if)#bandwidth 64	=	Router R2 R2(config)#interface serial 0/0/0 R2(config-if)#ip ospf cost 1562
R2(config)#interface serial 0/0/1 R2(config-if)#bandwidth 128	=	R2(config)#interface serial 0/0/1 R2(config-if)#ip ospf cost 781
Router R3 R3(config)#interface serial 0/0/0 R3(config-if)#bandwidth 256	=	Router R3 R3(config)#interface serial 0/0/0 R3(config-if)#ip ospf cost 390
R3(config)#interface serial 0/0/1 R3(config-if)#bandwidth 128	=	R3(config)#interface serial 0/0/0 R3(config-if)#ip ospf cost 781

OSPF and Multiaccess Networks

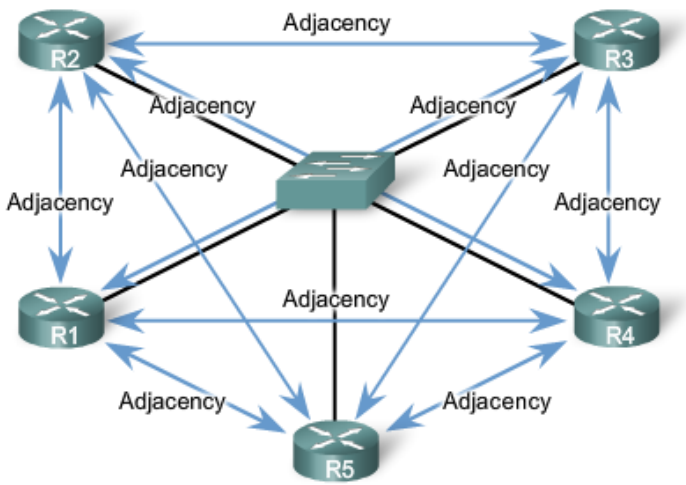
- Challenges in Multiaccess Networks
- OSPF defines five network types:
  - Point-to-point
  - Broadcast Multiaccess
  - Nonbroadcast Multiaccess (NBMA)
  - Point-to-multipoint
  - Virtual links



OSPF in Multiaccess Networks

- 2 challenges presented by multiaccess networks
  - Multiple adjacencies
  - Extensive LSA flooding

Number of Adjacencies Grows Exponentially

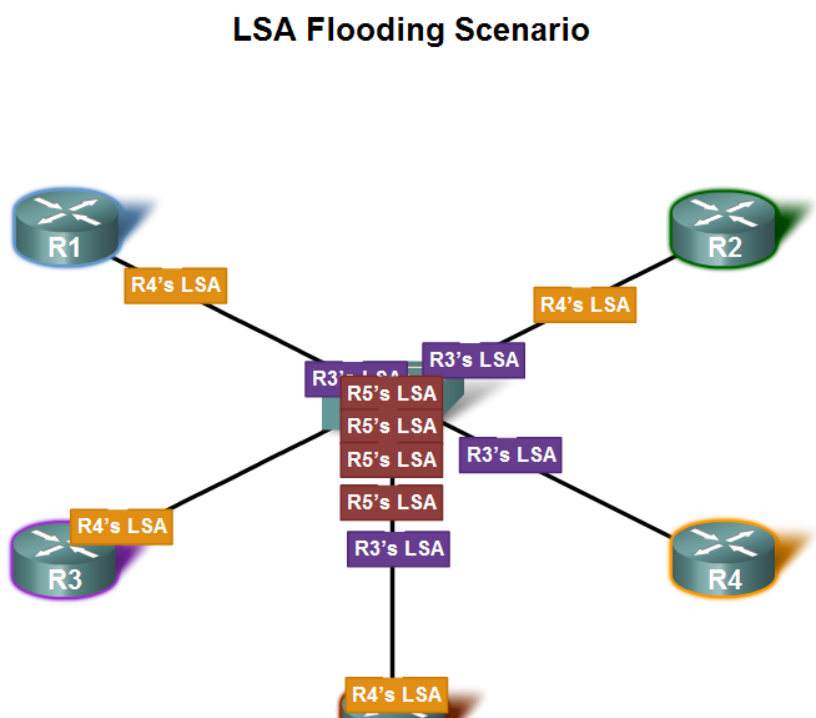


Routers	Adjacencies
n	n(n-1)/2
5	10
10	45
20	190
100	4,950

Number of Adjacencies = n(n-1)/2
n = number of routers
Example: 5 routers (5 - 1)/2 = 10 adjacencies

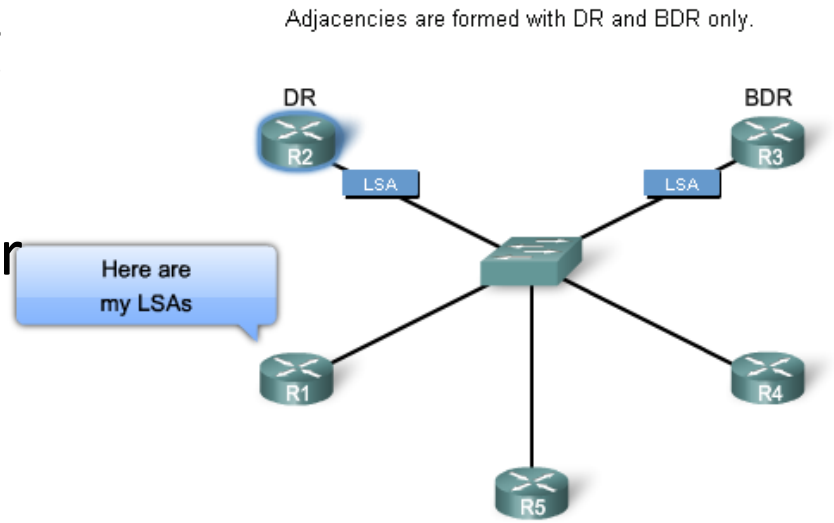
OSPF in Multiaccess Networks

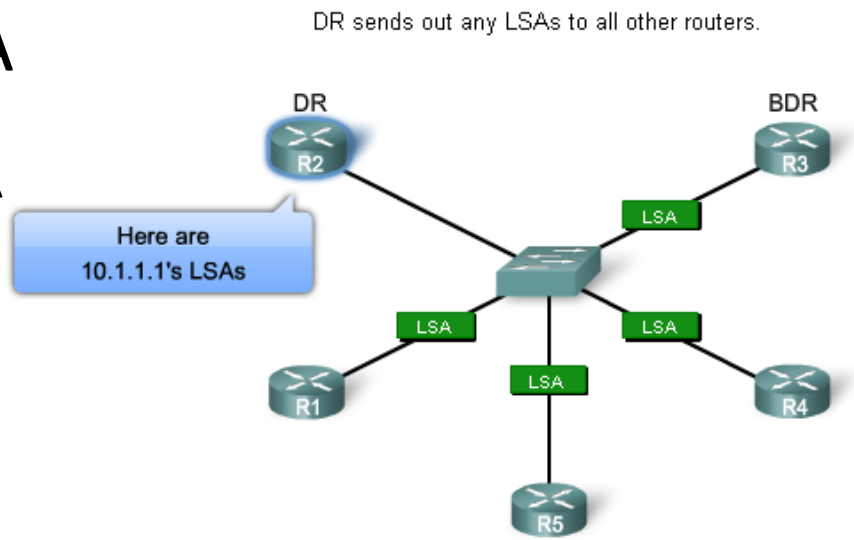
- Extensive flooding of LSAs
  - For every LSA sent out there must be an acknowledgement of receipt sent back to transmitting router
  - Consequence: lots of bandwidth consumed and chaotic traffic



OSPF in Multiaccess Networks

- Solution to LSA flooding issue is the use of
  - Designated router (DR)
  - Backup designated router (BDR)
- DR & BDR selection
  - Routers are elected to send & receive LSA
- Sending & Receiving LSA
  - DR others send LSAs via multicast 224.0.0.6 to DR & BDR
  - DR forward LSA via multicast address 224.0.0.5 to all other routers

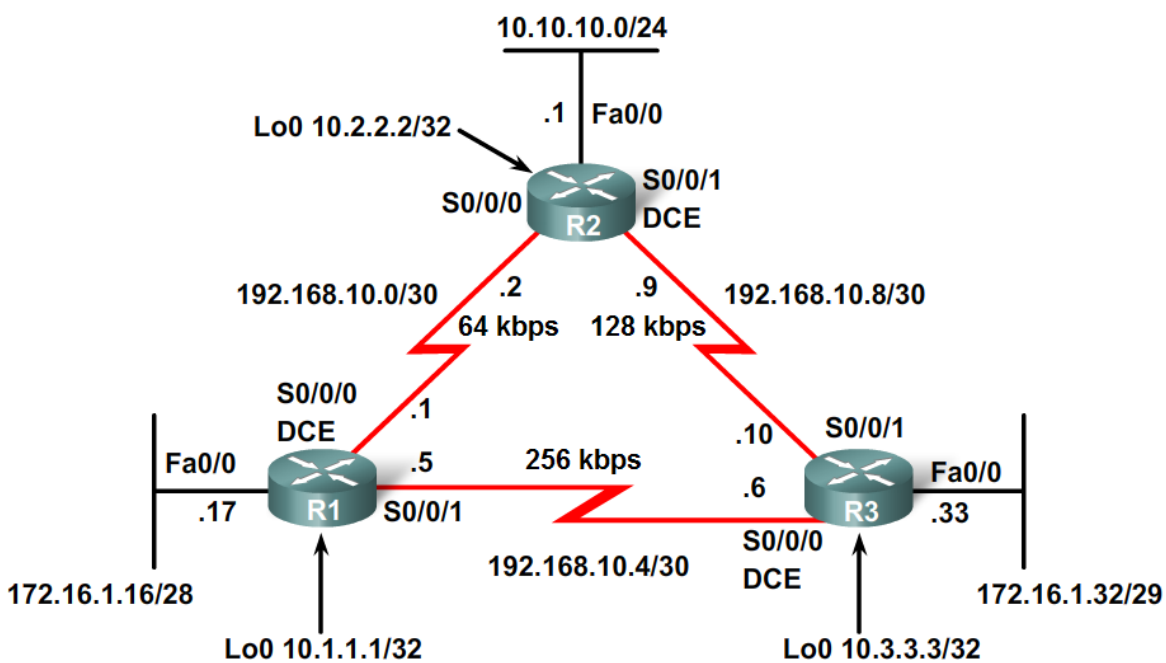




OSPF in Multiaccess Networks

- DR/BDR Election Process
  - DR/BDR elections **DO NOT** occur in point to point networks

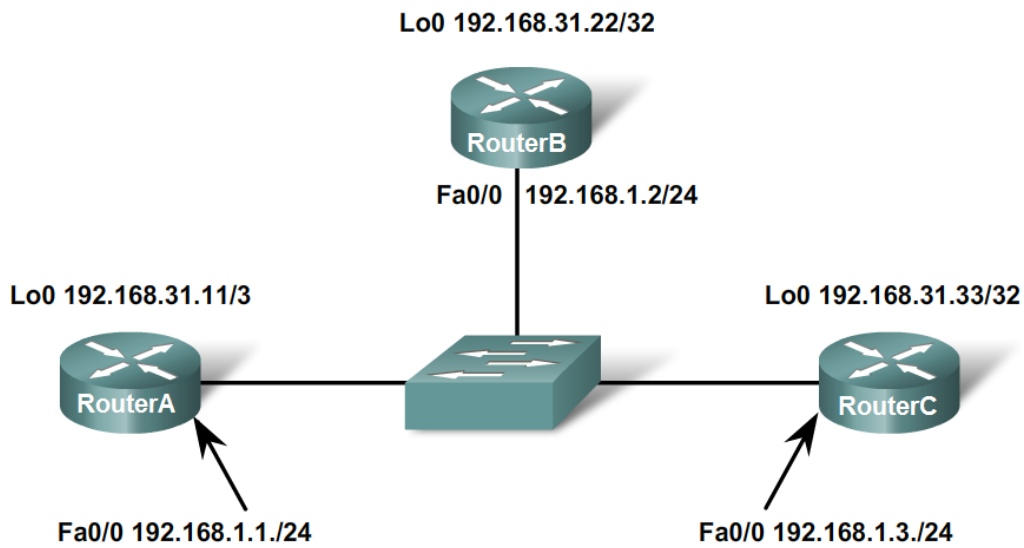
Point-to-Point Three Router Topology



OSPF in Multiaccess Networks

- DR/BDR elections **will take place on multiaccess networks** as shown below

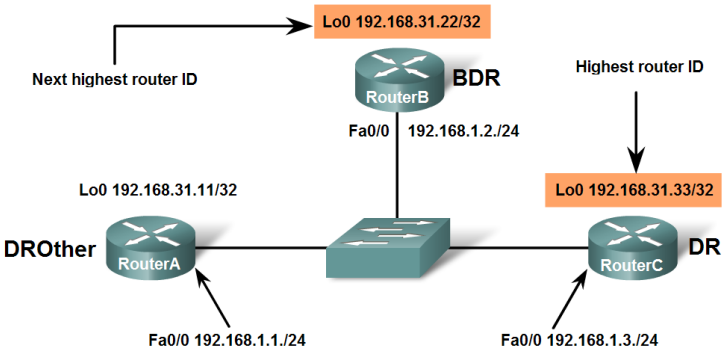
Multiaccess Three Router Topology



Notice that routers are now communicating via LAN interfaces.

OSPF in Multiaccess Networks

- Criteria for getting elected DR/BDR
  1. **DR:** Router with the highest OSPF interface priority
  2. **BDR**: Router with the second highest OSPF interface priority
  3. **If** OSPF interface priorities are equal, the highest router ID is used to break the tie




```
RouterA#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time    Address         Interface
192.168.31.33     1   FULL/DR         00:00:39     192.168.1.3     FastEthernet0/0
192.168.31.22     1   FULL/BDR        00:00:36     192.168.1.2     FastEthernet0/0
```

```
RouterB#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time    Address         Interface
192.168.31.33     1   FULL/DR         00:00:34     192.168.1.3     FastEthernet0/0
192.168.31.11     1   FULL/DROTHER    00:00:38     192.168.1.1     FastEthernet0/0
```

```
RouterC#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time    Address         Interface
192.168.31.22     1   FULL/BDR        00:00:35     192.168.1.2     FastEthernet0
192.168.31.11     1   FULL/DROTHER    00:00:32     192.168.1.1     FastEthernet0
```

Priority is equal at the default value of 1.

```
RouterA#show ip ospf interface fastethernet 0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.31.11, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 192.168.31.33, Interface address 192.168.1.3
  Backup Designated router (ID) 192.168.31.22, Interface address 192.168.1.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06
  Supports Link-local Signaling (LLS)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 192.168.31.22  (Backup Designated Router)
    Adjacent with neighbor 192.168.31.33  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

OSPF in Multiaccess Networks

- Timing of DR/BDR Election
  - Occurs as soon as 1st router has its interface enabled on multiaccess network
    - When a DR is elected it remains as the DR until one of the following occurs
      - The DR fails
      - The OSPF process on the DR fails
      - The multiaccess interface on the DR fails

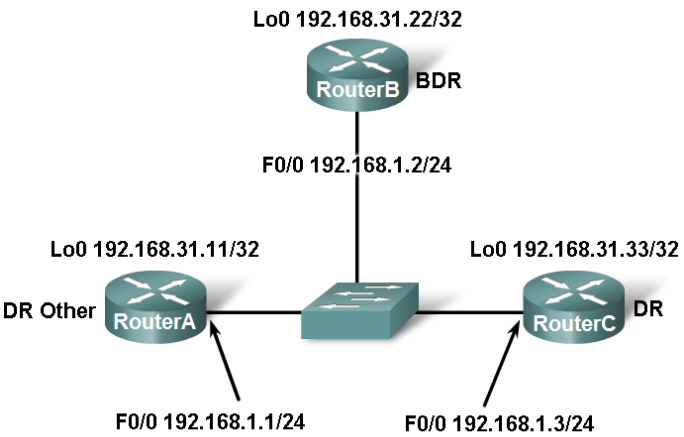
OSPF in Multiaccess Networks

- Manipulating the election process
  - If you want to influence the election of DR & BDR then do one of the following:
    - Boot up the DR first, followed by the BDR, and then boot all other routers

      OR
    - Shut down the interface on all routers, followed by a **no shutdown** on the DR, then the BDR, and then all other routers

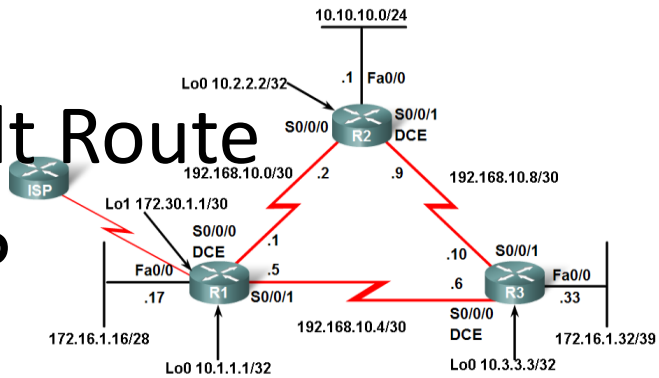
OSPF in Multiaccess Networks

- OSPF Interface Priority
- Manipulating the DR/BDR election process continued
  - Use the ip ospf priority interface command.
  - Example:Router(config-if)#**ip ospf priority** {**0 - 255**}
    - Priority number range 0 to 255
      - 0 means the router cannot become the DR or BDR
      - 1 is the default priority value




```
RouterA#show ip ospf interface fastethernet 0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.31.11, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 192.168.31.33, Interface address 192.168.1.3
  Backup Designated router (ID) 192.168.31.22, Interface address 192.168.
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
```

More OSPF Configuration



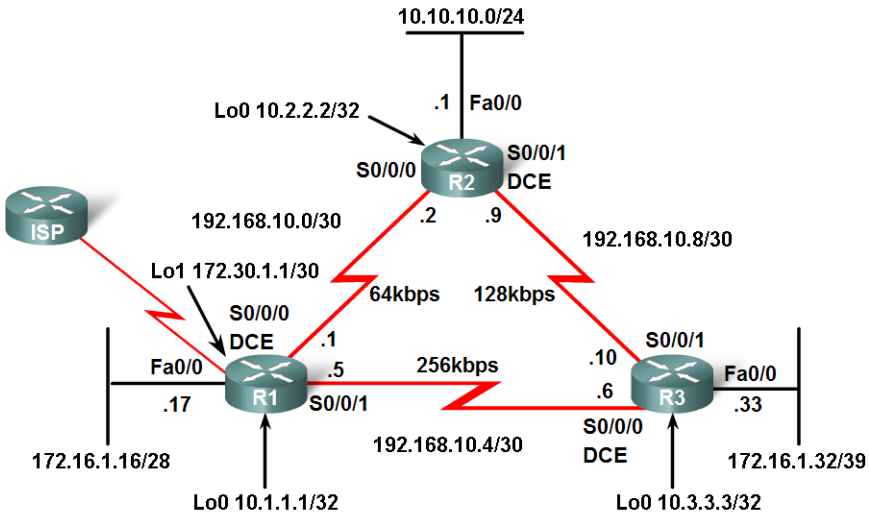
- Redistributing an OSPF Default Route
- Topology includes a link to ISP
  - Router connected to ISP
    - Called an autonomous system border router
    - Used to propagate a default route
      - Example of static default route:

        R1(config)#**ip route 0.0.0.0 0.0.0.0 loopback 1**
      - Requires the use of the **default-information originate** command
      - Example of default-information originate command:

        R1(config-router)#**default-information originate**

More OSPF Configuration

- Fine-Tuning OSPF
- Since link speeds are getting faster it may be necessary to change reference bandwidth values
  - Do this using the **auto-cost reference-bandwidth** command
  - Example:
    - R1(config-router)#auto-cost reference-bandwidth 10000




```
R1(config-if)#router ospf 1
R1(config-router)#auto-cost reference-bandwidth ?
  <1-4294967>  The reference bandwidth in terms of Mbits per second

R1(config-router)#auto-cost reference-bandwidth  10000
% OSPF: Reference bandwidth is changed.
```

```
R2(config-if)#router ospf 1
R2(config-router)#auto-cost reference-bandwidth  10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

```
R3(config-if)#router ospf 1
R3(config-router)#auto-cost reference-bandwidth  10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

More OSPF Configuration

- Fine-Tuning OSPF
- Modifying OSPF timers
  - Reason to modify timers
    - Faster detection of network failures
  - Manually modifying Hello & Dead intervals
    - Router(config-if)#**ip ospf hello-interval seconds**
    - Router(config-if)#**ip ospf dead-interval seconds**
  - Point to be made
    - Hello & Dead intervals must be the same between neigh

```
R1(config)#interface serial 0/0/0
R1(config-if)#ip ospf hello-interval 5
R1(config-if)#ip ospf dead-interval 20
R1(config-if)#end

<Wait 20 seconds for IOS message>
```

Summary

- RFC 2328 describes OSPF link state concepts and operations
- OSPF Characteristics
  - A commonly deployed link state routing protocol
  - Employs **DR**s & **BDR**s on multi-access networks
    - DRs & BDRs are elected
    - DR & BDRs are used to transmit and receive LSAs
  - Uses 5 packet types:

    1: HELLO
    2: DATABASE DESCRIPTION
    3: LINK STATE REQUEST
    4: LINK STATE UPDATE
    5: LINK STATE ACKNOWLEDGEMENT

Summary

- OSPF Characteristics
  - Metric = cost
    - Lowest cost = best path
- Configuration
  - Enable OSPF on a router using the following command
    - R1(config)#**router ospf *process-id***
  - Use the network command to define which interfaces will participate in a given OSPF process
    - Router(config-router)#**network** network-address wildcard-mask **area** area-id

Summary

- Verifying OSPF configuration
  - Use the following commands:
    - show ip protocol
    - show ip route
    - show ip ospf interface
    - show ip ospf neighbor

CISCO